Proposal: AI powered experiments on $SL_2(\mathbb{Z}_p)$

Benjamin Krawciw, Jake Quinn, Cody Strouse, Steph Tilneac, Wenzhuo Wu

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We propose using AI assisted programming to explore the diameter of the group $SL_2(\mathbb{Z}_p)$. Linear groups are ubiquitous in physics and mathematics. $SL_2(\mathbb{Z}_p)$ in particular is key in the study of representation theory, finite groups, and number theory.

$SL_2(\mathbb{Z}_p)$ has two generators, S and T. Given any γ in $SL_2(\mathbb{Z}_p)$ we can consider how many applications of S and T are needed to reach it. There is a smallest such number, call it $\ell(\gamma)$, and as $SL_2(\mathbb{Z}_p)$ is finite, there exists some γ in the group such that $\ell(\gamma)$ is greater than or equal to any other $\ell(\alpha), \alpha \in SL_2(\mathbb{Z}_p)$. This value is known as the diameter of the group, or informally, God's number. Denote God's number by $G(SL_2(\mathbb{Z}_p))$ [1].

It is known that $G(SL_2(\mathbb{Z}_p)) = O(\log(p))$, but the asymptotic constant C is not known. Also, God's numbers for $SL_2(\mathbb{Z}_p)$ are not usually computed or tabulated. By computing $G(SL_2(\mathbb{Z}_p))$ for many p values, we can find an experimental estimate for C. The Cayley graphs of $SL_2(\mathbb{Z}_p)$ are defined with the group elements as the vertices, and the edges are formed by applying S or T to reach another vertex. This means that the edges are directed and colored (by the choice of S or T). These will be Ramanujan graphs, which are known for their incredibly rapid growth [2]. By computing $G(SL_2(\mathbb{Z}_p))$, we are finding the radius of some of these graphs.

We are confident that we will be able to accomplish the following:

1. Implementation of Group Arithmetic: Develop a robust Python/C++ class for $SL(2, \mathbb{Z}/p\mathbb{Z})$ arithmetic that handles matrix multiplication and inversion efficiently modulo p.
2. Exact Diameter Calculation (Small p): Implement a Breadth-First Search (BFS) algorithm to compute the exact Cayley graph and God's number for small primes (e.g., $p = 3, 5, 7, 11, \ldots$ up to maybe $p \approx 100$).
3. Data Tabulation: Create a table of pairs $(p, G(p))$ and plot $G(p)$ against $\log(p)$.
4. Linear Regression: Perform a least-squares fit on the experimental data to estimate the constant C in the relationship $G(p) \approx C \log p$.

If time permits, we will also do the following:

1. AI/RL Solver Implementation: Train a Reinforcement Learning agent (e.g., Deep Q-Network or PPO, like in [3]) to find short paths for elements in groups with larger p (e.g., $p > 500$) where BFS is computationally intractable.

2. Heuristic Bound Estimation: Use the lengths found by the AI solver (OpenEvolve) as an upper bound for the diameter of larger groups. While AI cannot prove the true diameter (it might miss the optimal path), it provides a "practical" diameter.

3. Distribution Analysis: Instead of just the diameter (maximum), analyze the distribution of path lengths. Does the distribution of lengths converge to a specific shape as $p \to \infty$?

4. Generator Sensitivity (Optional): Briefly explore how changing the generating set changes the constant C.

References

[1] H. A. Helfgott. Growth and generation in sl_2(z/pz), 2008.

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